Propensity score matching for causal inference and treatment effects

What is PSM technique in causal, also can PSM be used for CATE and ATE measuments? apart from ATT

Propensity Score Matching (PSM)

PSM is a quasi-experimental technique for estimating causal effects from observational data when randomization isn't possible. The core idea, from Rosenbaum & Rubin (1983): instead of matching units on a high-dimensional vector of covariates X, you match on a single scalar — the **propensity score** $e(X) = P(T = 1 \mid X)$, the probability of receiving treatment given covariates.

The mechanics

1. **Estimate propensity scores** — fit a model (logistic regression, gradient boosting, etc.) predicting treatment assignment $T \in \{0, 1\}$ from observed covariates X.
2. **Match** treated units to control units with similar $e(X)$ — nearest neighbor, caliper, k:1 matching, kernel, stratification, or full matching.
3. **Check balance** — standardized mean differences (SMD < 0.1 is the usual threshold) on covariates between matched groups.
4. **Estimate the effect** on the matched sample, typically as a simple mean difference.

The key theoretical result: if unconfoundedness holds ($Y(0), Y(1) \perp T \mid X$), then it also holds conditional on $e(X)$ alone. That collapses the curse of dimensionality of exact matching.

Can PSM estimate ATE and CATE? (the actually interesting question)

ATT (Average Treatment effect on the Treated) is PSM's natural estimand. You match each treated unit to similar controls, average the differences. The estimand is $E[Y(1) - Y(0) \mid T = 1]$ — what was the effect *for those who actually got treated*.

ATE — yes, but with caveats. To get $E[Y(1) - Y(0)]$ over the whole population, you need counterfactuals for *both* groups. Options:

- **Symmetric matching**: match treated→control *and* control→treated, then average. Works but requires good overlap on both sides of the propensity distribution.
- **Full matching / optimal matching**: every unit gets matched to at least one from the other arm with weights. Stuart's work shows this is reasonable for ATE.

- **IPTW (Inverse Probability of Treatment Weighting)**: technically uses propensity scores but isn't "matching." Weight by $1/e(X)$ for treated and $1/(1-e(X))$ for controls. This is the more principled propensity-score route to ATE. Cleaner asymptotic properties, but variance explodes near $e(X) \in \{0, 1\}$ — hence clipping/trimming/stabilized weights.
- **Stratification on the propensity score** with within-stratum effects weighted by population proportion → ATE.

The honest answer: PSM *can* target ATE, but if ATE is what you want, IPTW or doubly-robust methods (AIPW, TMLE) are usually the better tools. PSM's comparative advantage is ATT under selection-on-observables.

CATE ($E[Y(1) - Y(0) \mid X = x]$) — this is where PSM gets weak. CATE requires modeling effect heterogeneity as a function of covariates, and matching discards information by collapsing X into a scalar. You *can* do CATE-via-PSM by matching within strata defined by effect modifiers, or by estimating ATT within subgroups, but it's clunky. For CATE in your context, the modern stack is much better suited:

- **Meta-learners**: S-learner, T-learner, **X-learner** (Künzel et al.), **R-learner** (Nie & Wager) — the R-learner is particularly elegant because it uses propensity scores as nuisance functions in a Robinson-style orthogonalization.
- **Causal Forests** (Athey, Tibshirani, Wager) — honest splitting on treatment-effect heterogeneity.
- **DR-learner / AIPW-based CATE estimators** — doubly robust and orthogonal.

These use propensity scores as a *component* (typically for IPW reweighting or orthogonalization) rather than as the matching key itself. Given your WTP/contextual bandit work, the R-learner framing should feel natural — it's essentially a partialling-out trick where you regress residualized outcomes on residualized treatment, weighted appropriately.

When PSM is the right tool vs. when it isn't

PSM shines when: you have a clear treated/control distinction, decent overlap, the ATT is the policy-relevant estimand, and you want something interpretable for stakeholders ("here's a matched cohort, here's the lift"). It's also reasonably defensible in regulated/lending contexts because the matched-cohort framing is easy to audit.

It struggles when: overlap is poor, treatment is continuous, you care about heterogeneity, or unconfoundedness is shaky (PSM does nothing about unobserved confounders — a common misconception). King & Nielsen's 2019 critique ("Why Propensity Scores Should Not Be Used for Matching") argues PSM can actually *increase* imbalance versus methods like Mahalanobis or coarsened exact matching in some regimes, because it discards information about which covariates matter. Worth reading if you're going to use PSM in production.

For your kind of problems — lending offer allocation, BBPS cost decisions — if you're doing observational causal inference, I'd lean toward DR/orthogonal methods (DML, causal forests) over vanilla PSM, and reserve PSM for cohort-comparison readouts where stakeholders need an interpretable "matched A/B-like" story.

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Can you help me understand this?

The propensity score theorem (Rosenbaum & Rubin, 1983)

This is one of the most elegant results in causal inference. Let me unpack it carefully.

Setting up the two conditions

Unconfoundedness on X (the assumption you start with):

$$Y(0), Y(1) \perp T \mid X$$

In plain English: *once you condition on all the covariates X, treatment assignment is as good as random.* There's no hidden confounder lurking — whatever made some units more likely to be treated is fully captured in X. Within any stratum of X, treated and control units are exchangeable.

Unconfoundedness on $e(X)$ (the consequence):

$$Y(0), Y(1) \perp T \mid e(X)$$

The claim: *the same property holds if you condition only on the scalar propensity score $e(X) = P(T = 1 \mid X)$.*

Why this is surprising

X might be 50-dimensional. The propensity score is **one number**. The theorem says: collapsing X down to this single scalar loses nothing for causal identification. You can match, stratify, or weight on $e(X)$ alone and recover the same causal effect you'd get by matching on the full X.

That should feel suspicious at first. We've thrown away enormous amounts of information about X — and yet conditional independence is preserved?

Why it actually works — the intuition

The propensity score is a **balancing score**. The defining property (proved in the same 1983 paper) is:

$$X \perp T \mid e(X)$$

Read that carefully: *among units with the same propensity score, the distribution of X is identical between treated and control groups.*

This is the crux. If two units have $e(X) = 0.3$, one treated and one control, they might have arrived at that 0.3 via very different covariate profiles — but **on average**, treated and control units sharing a propensity score have the same covariate distribution. The propensity score "absorbs" all the systematic differences in X that drive treatment assignment.

Once X is balanced across treatment groups within levels of $e(X)$, conditioning on $e(X)$ effectively conditions on X for the purpose of removing confounding bias. The covariates can still vary unit-to-unit, but they vary *the same way* in both arms.

A quick proof sketch

You want to show $P(T = 1 \mid Y(0), Y(1), e(X)) = P(T = 1 \mid e(X))$.

$$P(T = 1 \mid Y(0), Y(1), e(X)) = E[T \mid Y(0), Y(1), e(X)]$$

By the tower property, condition on X inside:

$$= E\big[E[T \mid Y(0), Y(1), X, e(X)] \,\big|\, Y(0), Y(1), e(X)\big]$$

Now use the original unconfoundedness assumption: $T \perp (Y(0), Y(1)) \mid X$, so $E[T \mid Y(0), Y(1), X] = E[T \mid X] = e(X)$. Note $e(X)$ is a function of X so it drops out of the conditioning set trivially:

$$= E[e(X) \mid Y(0), Y(1), e(X)] = e(X)$$

And $P(T = 1 \mid e(X)) = E[T \mid e(X)] = E[E[T \mid X] \mid e(X)] = E[e(X) \mid e(X)] = e(X)$.

Both sides equal $e(X)$. Done. ∎

Why "curse of dimensionality" matters here

Without this theorem, you'd need to match on the full X — find treated and control units with nearly identical covariate vectors. In high dimensions this is hopeless: exact matches don't exist, and "nearest neighbor" in 50-dimensional space is essentially random (everything is far from everything). Nonparametric estimators that condition on X converge at rate $n^{-2/(4+d)}$ — useless for $d = 20$.

The propensity score gives you a 1-dimensional sufficient statistic for confounding adjustment. You match on a scalar. Convergence rates become tractable. That's the dimensionality reduction the theorem buys you.

The catch worth internalizing

The theorem says conditioning on $e(X)$ is *sufficient* to remove confounding **given that you knew the true $e(X)$**. In practice you estimate $\hat{e}(X)$, and the quality of that estimate matters enormously. A misspecified propensity model means $\hat{e}(X)$ doesn't balance X — and then the magic evaporates. This is why balance diagnostics (SMDs after matching/weighting) are non-negotiable: they're how you check whether your *estimated* propensity score behaves like a real balancing score on your data.

It's also worth noting what the theorem does *not* say: it does not say PSM works without unconfoundedness. The original assumption $Y(0), Y(1) \perp T \mid X$ has to hold. If there's an unobserved confounder, no propensity score in the world fixes that — you've just compressed a flawed adjustment set into a scalar.

The takeaway: the propensity score is a dimensionality-reduction trick *conditional on selection-on-observables*. It makes a tractable estimator out of an intractable one, but it doesn't relax the identifying assumption.

For the calculation of ATT, actually the intuition is very clear. We have to take the treated items and match the control items in a way such that the propensity score matches are correct. Also, once we have scanned through the entirety of all the treated items in there and matched it up with the control items group, then we can measure the difference in the absolute change in the treatment, which may be the target variable itself, right, but for the control I am not able to understand what we have to scan and then move from an algorithmic point of view. Where it will be okay to use it and where it might not work well? What is a distribution that you are talking about, which, if it is not proper, then it will not work? Is the similar kind of a distribution problem that will exist with the case of ATT measurement as well?

ATT vs ATE: the algorithmic and distributional differences

Good question — let me work through this carefully because the asymmetry between ATT and ATE is where a lot of practitioner confusion lives.

ATT: the algorithm you already understand

You've got it right. The procedure:

1. Loop over every treated unit i with $T_i = 1$
2. Find one (or k) control units with similar $\hat{e}(X)$
3. Compute $Y_i - \bar{Y}_{\text{matched controls}}$
4. Average across all treated units

$$\widehat{\text{ATT}} = \frac{1}{N_1} \sum_{i:T_i=1} \left(Y_i - \frac{1}{k} \sum_{j \in \mathcal{M}(i)} Y_j \right)$$

The estimand answers: *for the units that actually got treated, what was the treatment effect?* The treated group **is** your reference population. Controls are merely a device to impute the missing counterfactual $Y_i(0)$ for treated units.

ATE: what changes algorithmically

ATE asks something different: *if we treated everyone in the population, vs treated no one, what would the average effect be?*

$$\text{ATE} = E[Y(1) - Y(0)]$$

This means you need counterfactuals for **both** groups:

- For treated units, you need $Y(0)$ — what would have happened without treatment
- For control units, you need $Y(1)$ — what would have happened with treatment

So the algorithm becomes symmetric:

1. **Forward pass**: for each treated unit i, find matched controls → estimate $Y_i(0)$
2. **Reverse pass**: for each control unit j, find matched treated units → estimate $Y_j(1)$
3. Compute unit-level effects: $\hat{\tau}_i = Y_i - \hat{Y}_i(0)$ for treated, $\hat{\tau}_j = \hat{Y}_j(1) - Y_j$ for controls
4. Average across **everyone**:

$$\widehat{\text{ATE}} = \frac{1}{N} \sum_{i=1}^{N} \hat{\tau}_i$$

This is the part you were asking about — yes, you scan the control group too, and for each control unit, you find treated units with similar propensity scores to impute what *would* have happened if that control had been treated.

The distribution issue — overlap (a.k.a. positivity, common support)

Here's where ATE gets much harder than ATT, and this is the "distribution problem" referenced earlier.

Overlap assumption: for the propensity score to work, you need $0 < e(X) < 1$ for all X in your population. Every unit must have *some* nonzero probability of being either treated or control.

In real data, the propensity distributions for treated and control look something like this:

```
Density
  |     /\ controls
  |    /  \          /\ treated
  |   /    \        /  \
  |  /      \      /    \
  | /        \   /       \
  └──────────────────────────────→ e(X)
  0                          1
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The two distributions overlap in the middle but have **thin tails** on opposite ends:

- Region near $e(X) \approx 0$: lots of controls, very few treated. Why? Because units with low propensity scores are units the model says "shouldn't get treated" — and indeed, few of them did.
- Region near $e(X) \approx 1$: lots of treated, very few controls. Same logic reversed.

Why this hits ATE harder than ATT

For ATT: you're matching treated → control. You need control units available wherever treated units sit on the propensity scale. The problematic region is high $e(X)$ — treated units with $e(X) = 0.95$ are hard to match because controls are rare there. But the low $e(X)$ region (lots of controls, few treated) doesn't bother you, because you never need to find controls *for* those rare treated units — there aren't any treated units there to begin with.

For ATE: you need matches in **both directions**. Now the low $e(X)$ region also becomes a problem — those abundant controls with $e(X) = 0.05$ need matched treated units, and there are almost none. You're trying to impute "what would happen if we treated a unit that essentially never gets treated in the data." That's an extrapolation, not a match.

This is why I said earlier that ATE requires good overlap on **both sides** of the propensity distribution, whereas ATT only requires good overlap on the treated side.

Concrete example — your lending context

Suppose treatment is "approved for premium credit line" and you want the effect on repayment.

- $e(X)$ depends on income, bureau score, tenure, etc.
- A high-income, high-bureau customer has $e(X) \approx 0.98$. Almost always approved.
- A subprime, thin-file customer has $e(X) \approx 0.02$. Almost never approved.

ATT asks: "Among customers we actually approved, what was the effect of approval?" — well-defined and estimable. The high-$e(X)$ approved customers need matched controls (rare but findable if overlap isn't terrible). Low-$e(X)$ approved customers are unusual but their controls are abundant.

ATE asks: "If we approved every customer, what would the average effect be?" — this requires answering "what would have happened if we'd approved the subprime thin-file customer?" That's an extrapolation into a region where the data is silent. You're asking the model to invent a counterfactual using essentially no comparable cases. The estimate becomes noisy and model-dependent.

Hence the practitioner rule of thumb: if overlap is poor, **redefine your estimand**. Either:

- Switch to ATT (only the treated population)
- Trim the sample to the region of common support (estimate ATE on the trimmed population — a different estimand sometimes called the ATO, average treatment effect on the overlap population)
- Use methods that handle weight instability better (stabilized IPW, overlap weights from Li, Morgan, Zaslavsky)

Where ATT can still fail distributionally

You asked whether ATT has a similar problem. Yes, but it's narrower.

ATT fails when: there are treated units in regions of X where **no comparable controls exist at all**. Imagine all the treated units with $e(X) > 0.9$ — if your control group has essentially zero density there, you can't impute their counterfactuals reliably. In matching, you'd end up either:

- Matching them to controls with very different propensity scores (bad matches, bias)
- Discarding them via caliper (now you're estimating ATT on a *subset* of the treated, not the full ATT)

The standard fix is **caliper matching with explicit reporting** of how many treated units got dropped, and acknowledging that you're now estimating ATT on the matched subset, not the originally defined ATT.

So the distributional issue exists for both, but:

- **ATT**: only requires controls to "cover" the treated propensity distribution
- **ATE**: requires both groups to cover each other's propensity distributions

Practical diagnostic

Before running PSM for ATE, always plot the propensity score histograms by treatment group. If the distributions barely overlap, ATE is not identified from your data — no algorithm fixes this. If they overlap heavily, you're fine. If overlap is decent in the middle but thin in tails, consider trimming or switching estimand.

For your work specifically: when stakeholders ask for "the effect of X" on a lending decision, they usually mean ATT (effect on the customers actually subjected to the policy), even if they phrase it as ATE. Push back and clarify the estimand — it usually saves you from chasing extrapolations that aren't defensible.

Understood to a large extent, but I have one question. Usually, if there is a bias which goes in the experimentation setup, then essentially, if I am only sending pns to users who are engaged for greater than 30 days with income greater than 10 lakh, something of that sort, and if those variables are also part of the estimation as well, meaning the X as well, then essentially it will be able to predict one with 100% certainty and zero with 100% certainty. Then essentially, the propensity or the treatment effect, the treatment scores that you are talking about, which is the core for psn that usually have either values lying in one or values lying in zeros, then essentially no test control is possible, right? Is my understanding right? In those kinds of situations, we can't identify the causal impact.

Yes, your understanding is exactly right — and you've just rediscovered the positivity violation problem

This is one of the most important practical insights in observational causal inference, and you've articulated it precisely. Let me sharpen the language and connect it to what's happening formally.

What you've described: deterministic treatment assignment

Your scenario: PNs sent only if `engagement_days > 30` AND `income > 10L`. Both variables are in X.

Then for any unit:

- If `engagement_days > 30` AND `income > 10L`: $e(X) = 1$ (always treated)
- Otherwise: $e(X) = 0$ (never treated)

There's no X in your data where treatment is *probabilistic*. It's deterministic. And this kills causal identification — completely.

Why it breaks PSM specifically

PSM needs to match treated and control units with similar $e(X)$. But:

- Among units with $e(X) = 1$: all are treated. Zero controls to match against.
- Among units with $e(X) = 0$: all are controls. Zero treated to match against.

There is no region of common support. The treated and control propensity distributions live on disjoint points (literally $\{0\}$ and $\{1\}$). Matching is mathematically undefined.

Why it breaks *every* method, not just PSM

This is the key generalization. The issue isn't PSM-specific — it's that the causal effect is **not identified** from this data, period. Other methods just fail in different ways:

- **IPW**: weights are $1/e(X)$ for treated and $1/(1 - e(X))$ for controls. With $e(X) \in \{0, 1\}$, you get division by zero. Weights blow up to infinity.
- **Regression adjustment / S-learner**: you fit $E[Y \mid X, T]$. But for units with `engagement_days > 30` AND `income > 10L`, you have *only* $T = 1$ observations — the model has no data to learn $E[Y \mid X, T = 0]$ in that region. It will extrapolate using functional form assumptions (linearity, etc.), and those assumptions are doing 100% of the work. You're not estimating a causal effect, you're trusting a parametric extrapolation.
- **Doubly robust methods (AIPW, TMLE)**: doubly robust means "consistent if *either* the outcome model *or* the propensity model is correct." But here, both nuisance models fail in the same region for the same reason. Double robustness doesn't save you from positivity violations — it only protects against misspecification when positivity holds.
- **Causal forests, X-learner, R-learner**: same problem. They need variation in T within neighborhoods of X to estimate heterogeneous effects. Deterministic assignment means no variation. The tree can't split usefully, the meta-learner has no counterfactual signal.

The formal name: positivity (a.k.a. overlap, common support)

This is the **positivity assumption**, one of the three core identifying assumptions for causal inference from observational data (alongside unconfoundedness and SUTVA/consistency):

$$0 < P(T = 1 \mid X = x) < 1 \quad \text{for all } x \text{ with } P(X = x) > 0$$

In words: every unit in the population must have a nonzero probability of being in *either* treatment arm. Your PN setup violates this — it forces $P(T = 1 \mid X) \in \{0, 1\}$ over the support of X.

When positivity fails, the causal estimand is **not identifiable**. No estimator, however clever, can recover it from this data alone. This is a property of the data-generating process, not the algorithm.

The intuition in one sentence

Causal inference from observational data requires that *for any kind of person, you can observe both what happens when they're treated and what happens when they're not*. If your policy guarantees that high-income engaged users are *always* treated, then your data simply contains no information about what would happen to such users if they weren't treated. No statistical technique can manufacture that information.

Practical implications for your work

This is enormously relevant to lending and recommendation contexts because **policy-driven treatment assignment is everywhere**. Some examples:

- "Premium credit line offered only to bureau score > 750" → can't estimate effect of premium line for those customers using observational data
- "Promotional rate only for tenure > 6 months" → can't separate effect of promotional rate from effect of tenure on outcomes
- "Cashback campaign sent only to inactive users" → can't estimate counterfactual for active users

What you can do when positivity fails

Option 1: Redefine the estimand to the overlap region. Restrict analysis to the subpopulation where $0 < e(X) < 1$ actually holds — i.e., units where the policy *was* probabilistic in practice. Often there's some noise in real-world rule application (manual overrides, system errors, A/B holdouts) that creates a sliver of overlap. Estimate the effect on that subpopulation and be honest that it doesn't generalize to the deterministic regions.

Option 2: Drop the deterministic covariates from X — but only if defensible. If you exclude `engagement_days` and `income` from the propensity model, you'll get propensity scores strictly between 0 and 1 again. But this only works if those variables aren't confounders — i.e., they don't affect the outcome except through treatment. In your example they almost certainly *do* affect the outcome directly (engaged high-income users behave differently regardless of PNs), so dropping them reintroduces confounding. Trading positivity violation for unconfoundedness violation — usually a bad trade.

Option 3: Find a natural experiment. Look for exogenous variation in treatment assignment within the deterministic region. Did the rule change at some point? Are there A/B tests, randomized holdouts, system glitches, capacity constraints that created randomness in who got the PN? This is the cleanest path — it gives you a local experiment within the otherwise-deterministic regime.

Option 4: Run an actual experiment. The most reliable answer when positivity fails in observational data is to inject randomization deliberately. Hold out a small fraction of eligible users (say 5%) who would have gotten the PN under the rule but don't. This creates a controlled overlap region where causal effects are cleanly identified. Combined with the bulk observational data, this is what makes contextual bandit / off-policy evaluation frameworks tractable — which connects directly to your bandit work.

Option 5: Lean on structural assumptions and be explicit. If you must answer the question with the data you have, you can use parametric models that extrapolate (linear regression, structural models). But you're no longer doing causal inference from data — you're doing causal inference from data *plus* strong functional form assumptions. The honest version of this is to report the result alongside a sensitivity analysis showing how much it depends on the extrapolation.

The deeper takeaway

What you've identified is exactly why "we have observational data, let's just run PSM" is often a non-starter in production ML systems. The systems are usually rule-based or model-based with deterministic thresholds, which means positivity is violated almost by construction. The discipline that production data scientists need to develop is recognizing this *before* running any causal analysis: plot $\hat{e}(X)$ histograms first, look for the bimodal-at-0-and-1 pattern, and if you see it, the answer is either redesign the data collection (inject randomization) or redefine the question to the overlap subpopulation. No amount of methodological sophistication fixes a violated positivity assumption.

Your instinct here is exactly the kind of skepticism that separates people who use causal methods correctly from those who run the algorithm and trust the number it spits out.

Also, while calculating the PSM, will this PSM need to be calculated on a separate data which is outside of the modeling, or can I use the modeling data which I am using for doing the training?

Propensity model data: a subtle but important question

This is a question that trips up a lot of practitioners because it conflates two different concerns: (a) train/test splits for predictive ML, and (b) the role of the propensity model in causal inference. They follow different logic.

The short answer

The propensity model is typically fit on the **same data** you use for the causal effect estimation — not a separate holdout. This sounds wrong if you're coming from a predictive ML mindset (where "fit nuisance model and estimator on same data" screams overfitting), but for traditional PSM, it's standard practice and statistically defensible. *However*, modern doubly-robust and machine-learning-based methods (DML, AIPW with flexible nuisance learners) explicitly require **cross-fitting**, which is a more nuanced version of "separate data." Let me unpack both regimes.

Traditional PSM: same data is fine

In the classic Rosenbaum-Rubin setup with parametric propensity models (logistic regression), you:

1. Fit $\hat{e}(X)$ on the full dataset $\{(X_i, T_i)\}_{i=1}^N$
2. Compute matched pairs using $\hat{e}(X)$
3. Estimate ATT/ATE on the same data using the matched sample

Why this doesn't cause the kind of bias you'd worry about in predictive ML:

The propensity model isn't trying to predict Y. It's predicting T from X. The outcome Y doesn't enter the propensity model at all. So even if $\hat{e}(X)$ overfits T, that overfitting doesn't directly bias the relationship between treatment and outcome — it just produces a noisier matching key.

The estimand is the treatment effect, not predictive accuracy. You're not asking "how well does $\hat{e}(X)$ predict T on new data?" — you're asking "does conditioning on $\hat{e}(X)$ balance X across treatment groups in *this* sample?" That's a question you answer with balance diagnostics on this same sample, not via out-of-sample evaluation.

There's a subtle theoretical point: Abadie & Imbens (2016) showed that estimating $\hat{e}(X)$ rather than knowing the true $e(X)$ can actually *reduce* the variance of the matching estimator in finite samples, because the estimated score absorbs some of the sample-specific covariate imbalance. So using estimated scores on the same data isn't just acceptable — it can be statistically preferable to using oracle scores.

When same-data fitting *does* become a problem

The concern arises when the propensity model is **flexible and high-capacity** — gradient boosting, random forests, neural networks. These can overfit aggressively, and the failure mode is specific:

A flexible model can find an $\hat{e}(X)$ that perfectly separates treated and control units in your training data, even when the true $e(X)$ has substantial overlap. The propensity distribution histogram will look bimodal at 0 and 1 — not because positivity is actually violated, but because your model memorized the assignment. You'll then either:

- Discard most of the sample as "non-overlapping" (losing power)
- Get matched pairs with extreme estimated propensities (high variance, biased)
- Get IPW weights that explode

This is overfitting masquerading as positivity violation. The propensity scores are spurious — they don't reflect the true balancing structure, just the model's ability to memorize.

The fix is **cross-fitting** (sometimes called sample splitting or DML-style estimation).

Cross-fitting: the modern best practice

Chernozhukov et al. (2018) — Double/Debiased Machine Learning — formalized this. The procedure:

1. Split data into K folds (typically $K = 5$ or $K = 10$)
2. For each fold k:
   - Fit the propensity model on the *other* $K - 1$ folds
   - Predict $\hat{e}(X_i)$ for units in fold k using that out-of-fold model
3. Stack predictions across folds → you now have an $\hat{e}(X_i)$ for every unit, each computed from a model that never saw that unit during training
4. Use these out-of-fold propensity scores in your matching/weighting/effect estimation

This eliminates the overfitting bias. The estimated propensities behave like genuine predictions of treatment probability, not memorizations of treatment assignment. Theoretically, this restores the $\sqrt{n}$-convergence and asymptotic normality of the causal effect estimator even when using complex ML for the nuisance functions.

There's a refinement called **cross-fitting with K folds repeated multiple times** (averaging across different fold splits to reduce variance from the random split itself), which Chernozhukov's paper recommends for production use.

So which regime applies to you?

A practical decision rule:

Use same-data fitting if:

- Your propensity model is logistic regression with a modest feature set
- You're doing classical PSM with balance diagnostics
- Sample size is large relative to feature dimensionality
- You verify the propensity distribution looks reasonable (overlap, not bimodal-at-0-and-1)

Use cross-fitting if:

- Your propensity model is gradient boosting / random forest / neural net
- You're doing DML, AIPW, TMLE, or any doubly-robust method
- You're estimating CATE with meta-learners (R-learner, X-learner, DR-learner — these *require* cross-fitting for the orthogonality theory to hold)
- High-dimensional X or n/p ratio is not generous
- You're publishing results or using them for high-stakes decisions where variance estimation matters

What about train/test for the outcome model?

If you're also fitting an outcome model $\hat{\mu}(X, T) = E[Y \mid X, T]$ (for DR estimation, AIPW, or meta-learners), the same logic applies — and more strongly. The outcome model directly uses Y, so overfitting bias propagates straight into the effect estimate. Cross-fitting is essentially mandatory for outcome models in any flexible-ML causal inference pipeline.

The R-learner (Nie & Wager) makes this very explicit: you cross-fit the propensity model $\hat{e}(X)$ and the marginal outcome model $\hat{m}(X) = E[Y \mid X]$, then use the cross-fitted residuals in the final stage. The whole orthogonalization argument depends on the nuisance functions being computed out-of-fold.

Connecting to your modeling workflow

If you're building, say, a P(install) × P(KPI | install) cascade where you also want to assess the causal effect of some intervention (e.g., showing a different creative), the clean separation looks like:

- **Modeling data**: used to train the cascade itself

- **Propensity model for causal analysis**: fit via cross-fitting on whatever subset you're using for the causal study — could overlap with modeling data, but use cross-fitting so units are predicted by models that didn't train on them
- **Outcome model for DR estimation**: same — cross-fitted on the causal analysis sample

The cross-fitting *replaces* the need for a separate holdout. You don't need a third dataset. The K-fold structure within your causal analysis sample provides the necessary independence between nuisance estimation and effect estimation.

One genuine concern is **leakage between the predictive modeling and the causal analysis** if they share data. If your cascade is being deployed and influencing future treatment assignment in ways that show up in X, you can get circular dependencies. But that's a sequential / off-policy evaluation problem, distinct from the basic question of whether to use a holdout for the propensity score itself.

Practitioner-level summary

For classical logistic-regression PSM: same data is fine, just check balance.

For anything fancier: cross-fit. It's cheap (5-10x model fits, no big deal), it eliminates a real source of bias, and it's the standard that any reviewer or principal-level audit will expect to see. If you're using EconML, DoubleML, or CausalML libraries, cross-fitting is built in by default — you just specify `cv=5` and it handles the fold structure for you.

The mental model to internalize: **the propensity model is a nuisance function, and nuisance functions in flexible-ML causal inference must be cross-fitted to preserve the statistical properties of the effect estimator.** That's the single most important methodological upgrade from "vanilla PSM" to "modern causal ML."

what is IPTW?

IPTW: Inverse Probability of Treatment Weighting

IPTW is the propensity-score-based alternative to matching. Same input ($\hat{e}(X)$), same identifying assumptions (unconfoundedness + positivity), but a fundamentally different way of using the propensity score to estimate causal effects.

The core idea

Instead of *matching* treated units to similar controls and discarding the rest, IPTW *reweights* the entire sample so that the weighted treated group and weighted control group look like the same population — specifically, the full population you care about.

The weights:

$$w_i = \begin{cases} \dfrac{1}{\hat{e}(X_i)} & \text{if } T_i = 1 \\ \dfrac{1}{1 - \hat{e}(X_i)} & \text{if } T_i = 0 \end{cases}$$

Then the ATE estimator is just a weighted difference in means:

$$\widehat{\text{ATE}}_{\text{IPTW}} = \frac{1}{N} \sum_{i=1}^{N} \frac{T_i Y_i}{\hat{e}(X_i)} - \frac{1}{N} \sum_{i=1}^{N} \frac{(1 - T_i) Y_i}{1 - \hat{e}(X_i)}$$

That's it. No matching, no caliper, no fold structure for pairs. You compute propensity scores, build weights, take a weighted mean difference.

Why the weights work — the intuition

Think of it as **upweighting underrepresented units in each treatment arm**.

Suppose a unit has X such that $e(X) = 0.1$ — they're unlikely to be treated. If you happen to observe this unit as treated, they're rare in the treated group relative to how often this type of X appears in the population. So you upweight them by $1/0.1 = 10$ to compensate. Their voice gets amplified in the weighted treated mean.

Conversely, that same unit if observed as control has weight $1/(1 - 0.1) = 1.11$ — they're abundant in the control group already, no need to amplify.

The result: after weighting, the distribution of X in the treated arm matches the distribution of X in the full population. Same for the control arm. The two weighted arms become comparable to a synthetic randomized experiment.

The Horvitz-Thompson connection

This is actually an old idea — Horvitz & Thompson (1952) developed inverse-probability weighting for survey sampling, decades before causal inference adopted it. The logic is identical: if you sample units with unequal probabilities, you weight by the inverse of the sampling probability to recover population-level estimates. Treatment assignment in observational studies is essentially "sampling into the treated arm" with non-uniform probabilities — so the same correction applies.

Different estimands, different weights

ATE (effect on whole population): the weights above.

ATT (effect on the treated):

$$w_i^{\text{ATT}} = \begin{cases} 1 & \text{if } T_i = 1 \\ \dfrac{\hat{e}(X_i)}{1 - \hat{e}(X_i)} & \text{if } T_i = 0 \end{cases}$$

Treated units keep weight 1 (they already represent the target population). Controls get reweighted to *look like* the treated distribution of X.

ATC (effect on the controls — useful in policy rollout scenarios where you want to know "what would happen if we treated the currently untreated?"): flip the ATT logic.

IPTW vs PSM — when to use which

This is the practical question and worth thinking through carefully.

Advantages of IPTW over matching:

- **Uses all the data.** Matching discards unmatched units; IPTW keeps everyone (with appropriate weight).
- **Naturally extends to ATE.** No "forward and reverse pass" gymnastics — the same formula gives you ATE directly.
- **Handles continuous treatments.** Matching doesn't generalize well to continuous T; IPTW does (with generalized propensity scores).
- **Plays nicely with regression adjustment.** You can fit a weighted outcome regression on top, which gives you doubly robust estimation almost for free.
- **Variance is analytically tractable.** Sandwich estimators, GEE-style standard errors — well-developed machinery.

Disadvantages of IPTW:

- **Weight instability near positivity violations.** If $\hat{e}(X) \approx 0$ or ≈ 1 for some units, their weights explode — a single unit can dominate the entire estimate. This is IPTW's Achilles heel and the reason most production-grade implementations involve weight stabilization or trimming.
- **Less interpretable.** "I matched 1,247 treated customers to similar controls" is a story a product manager can follow. "I weighted each customer by the reciprocal of their estimated treatment probability" is not.
- **Sensitive to propensity model misspecification.** Matching is somewhat forgiving — small errors in $\hat{e}(X)$ produce small errors in match quality. IPTW amplifies propensity errors through the reciprocal, especially in the tails.

The weight explosion problem and fixes

This is where IPTW gets practically tricky. If a unit has $\hat{e}(X) = 0.01$ and shows up as treated, their weight is 100. If another has $\hat{e}(X) = 0.001$, weight is 1000. These extreme weights produce estimators with enormous variance — a single observation can swing the ATE estimate by huge amounts.

Standard remedies:

Trimming (truncation). Cap weights at some threshold (e.g., 1st/99th percentile, or fixed values like 10 or 20). Introduces a small bias but dramatically reduces variance. The bias-variance tradeoff is usually worth it.

Stabilized weights (Robins et al., 2000). Multiply by the marginal probability of the observed treatment:

$$w_i^{\text{stab}} = \begin{cases} \dfrac{P(T=1)}{\hat{e}(X_i)} & \text{if } T_i = 1 \\ \dfrac{P(T=0)}{1-\hat{e}(X_i)} & \text{if } T_i = 0 \end{cases}$$

This makes the weights cluster around 1 rather than ranging wildly, while preserving the inverse-probability logic. Recommended default in most modern applications.

Overlap weights (Li, Morgan, Zaslavsky, 2018). A more elegant approach: instead of $1/\hat{e}$ and $1/(1-\hat{e})$, use $1-\hat{e}$ for treated and $\hat{e}$ for controls. These weights are bounded in $[0,1]$, can never explode, and have the nice property that they implicitly target the *overlap population* — the subset of units with realistic probabilities of either treatment. This sidesteps positivity problems entirely by redefining the estimand to where the data can support it. Increasingly popular for production use.

The doubly robust bridge: AIPW

IPTW alone relies entirely on the propensity model being correct. Regression adjustment alone relies entirely on the outcome model being correct. **Augmented IPW (AIPW)** combines them:

$$\widehat{\text{ATE}}_{\text{AIPW}} = \frac{1}{N}\sum_{i=1}^{N}\left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{T_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-T_i)(Y_i - \hat{\mu}_0(X_i))}{1-\hat{e}(X_i)}\right]$$

The doubly robust property: this estimator is consistent if *either* $\hat{e}(X)$ is correct *or* $\hat{\mu}_t(X)$ is correct. You only need one of the two models right.

This is the modern default for ATE estimation when you have flexibility to model both. Combined with cross-fitting, it gives you the DML/AIPW estimator that's the workhorse of contemporary causal ML — the foundation that EconML, DoubleML, and similar libraries are built on.

Where IPTW fits in your toolkit

Given your work, the practical hierarchy:

1. **Use IPTW (stabilized, with trimming or overlap weights) when** you want ATE on a well-defined target population, overlap is reasonable, and you want to use all the data.
2. **Use matching when** you want an interpretable cohort-comparison narrative for stakeholders, ATT is the relevant estimand, or you want a simple readout that's easy to audit.
3. **Use AIPW / DML when** you have budget for both propensity and outcome modeling, want statistical efficiency, and need doubly robust protection against nuisance misspecification.
4. **Use causal forests / R-learner when** CATE is the goal — but note that under the hood, these methods use propensity scores via IPTW-style weighting in their orthogonalization steps. IPTW is the conceptual primitive.

The thread connecting all of this: **IPTW is not really a competitor to matching — it's a more general framework that subsumes matching as a special case** (matching is essentially a 0/1 weighting scheme based on neighborhood membership). Once you internalize the IPTW lens, matching starts to look like a restricted, less efficient version of the same idea, and the modern causal ML stack becomes a natural extension rather than a separate world.